

FIXED-WIDTH ESTIMATION OF THE COMMON MEAN OF A SET OF NORMAL POPULATIONS WITH UNKNOWN AND UNEQUAL VARIANCES

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Summary:

A sequential procedure is proposed for the construction of a fixed-width confidence interval for the common mean of several normal populations. The procedure first uses a method for the selection of a population with variance close to the smallest variance. Further observations are then taken only from the selected population. Asymptotic properties of the procedure are investigated. The procedure is then compared with certain existing two-stage procedures and asymptotic efficiencies are considered. It is shown that the proposed sequential procedure is asymptotically more efficient.

1. INTRODUCTION

Consider k populations $\pi_1, \dots, \pi_k$ where π_i is $N(\mu, \theta_i)$, $i = 1, \dots, k$. Here the common mean μ is unknown and the variances $\theta_1, \dots, \theta_k$ are unknown and unequal. The problem is to construct a confidence interval of length at most $2d$ for μ . The proposed solution is a sequential procedure which achieves the desired coverage probability in an asymptotic sense, to be made precise at a later stage.

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The problem arises in practice, for example, when an unknown quantity μ has to be estimated and it can be measured by k different instruments whose precisions are not known in advance.

Fixed sample point estimation of μ has been considered by Graybill and Deal (1959). The Graybill-Deal estimator is defined as the weighted mean of the sample averages with weights as reciprocals of the estimators of the variances of the sample averages. This estimator, variants of it and related confidence intervals have been extensively studied in the literature. See Sinha (1985) for references.

In the case of *two* normal populations, various sequential estimation procedures have been studied. Mallik (1971) has considered risk-efficient estimation of the common mean when one of the variances is known. Richter (1960) studied a similar problem and gave an asymptotic minimax two-stage procedure. (See also Alam (1967).) Mallik's (1971) solution is a combination of the solutions of two related problems, namely the sequential estimation of the mean of a normal population when the variance is unknown (considered by Robbins (1959) and Starr (1966)) and the one-armed bandit problem (considered by Chernoff (1968)). Chernoff (1971), using the solution of the two-armed bandit problem, solved the risk-efficient estimation problem when both variances are unknown. Govindarajulu (1987) contains a discussion of some of these procedures.

Alam and Saxena (1984) discuss two-stage procedures in order to obtain bounded length confidence intervals for μ . These procedures take an equal number of observations from each population in the first stage. On the basis of these observations it is decided from which population to sample in the second stage and the (random) sample size of the second stage is determined. Since the estimators of the population variances used in the intervals are based on the data in the first stage only, it is to be anticipated that a sequential procedure should be more efficient.

In section 2 we propose a sequential procedure. The procedure makes use of a method for selecting a population with variance close to the smallest variance, thus linking in an interesting way the theory of selection proce-

dures with that of sequential fixed-width confidence intervals. Section 3 contains an asymptotic analysis of the proposed procedure. In Section 4 the sequential procedure is compared with the two-stage procedures mentioned above, while Section 5 contains a discussion.

2. A SEQUENTIAL PROCEDURE BASED ON A METHOD FOR SELECTION OF THE SMALLEST VARIANCE

We first consider the fixed sample size case. Suppose n_i observations are available from the i 'th of k populations ($i = 1, \dots, k$) and let $X_{ij}, j = 1, \dots, n_i$, denote the observations from the $N(\mu, \theta_i)$ distribution. Let $\bar{X}_i(n_i) = n_i^{-1} \sum_{j=1}^{n_i} X_{ij}$ and $S_i(n_i) = n_i^{-1} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i(n_i))^2$. Then it is easily seen that the maximum likelihood estimators of μ and θ_i are given by

$$(2.1) \quad \hat{\mu}(n) = \sum_{i=1}^k n_i \bar{X}_i(n_i) \hat{\theta}_i(n_i)^{-1} / \sum_{i=1}^k n_i \hat{\theta}_i(n_i)^{-1}$$

$$(2.2) \quad \hat{\theta}_i(n_i) = n_i^{-1} \sum_{j=1}^{n_i} (X_{ij} - \hat{\mu}(n))^2 = S_i(n_i) + (\bar{X}_i(n_i) - \hat{\mu}(n))^2$$

where $n = \max_{1 \leq i \leq k} n_i$ (the dependence on n is used in the definition of the sequential procedure).

Note that $\hat{\mu}(n)$ and $\hat{\theta}_i(n_i) (i = 1, \dots, k)$ are defined implicitly in the above equations. However solutions may be obtained iteratively as follows: first substitute $\hat{\theta}_i(n_i)$ in (2.1) by $S_i(n_i)$ and obtain $\hat{\mu}_1(n)$. Now use $\hat{\mu}_1(n_i)$ in (2.2) and iterate until the desired accuracy is obtained.

An alternative to $\hat{\mu}$ is obtained by replacing $\hat{\theta}_i(n_i)$ by $S_i(n_i)$ in (2.1). The estimator obtained in this way is unbiased best asymptotically normal and has been widely studied. In the asymptotic analysis which follows it will usually make no difference which of these weights is used in (2.1).

In the construction of a sequential procedure it is desirable that the population with the smallest variance be identified at an early stage. Then further sampling can be confined to the selected population. Of course, if the variances were known, one would sample from the population with the smallest variance only. Since the variances are usually unknown in practice, one is led to the application of a selection procedure for first selecting the population with the smallest variance. Such procedures are available in the literature. We refer specifically to the procedure of Bechhofer and Sobel (1954) where an initial sample of predetermined size r is drawn from each population and the one with the smallest sample variance is selected. Tables for the implementation of the procedure are provided by Bechhofer and Sobel (1954). See also Table G in Gibbons, Olkin and Sobel (1977). This procedure uses the so-called indifference zone formulation, which may be stated as follows: Let $\theta_\lambda = \min(\theta_1, \dots, \theta_k)$, let $\Delta^* > 1$ and let ℓ be the index of the selected population, then we require

$$(2.3) \quad P(\ell = \lambda) \geq P^* \text{ whenever } \theta_i/\theta_\lambda \geq \Delta^* \text{ for all } i \neq \lambda, i = 1, \dots, k.$$

The value of r is determined, for given Δ^* and P^* , in such a way that (2.3) is satisfied.

It is, however, possible to make a stronger statement than (2.3) for this procedure (cf. Parnes and Srinivasan (1986)). The stronger statement, which will be useful in our procedure, is as follows: For the same value of r which guarantees (2.3), we have

$$(2.4) \quad P(\theta_\ell/\theta_\lambda < \Delta^*) \geq P^*$$

Note that consideration of (2.3) alone, would mean that probability statements concerning the procedure to be constructed hold true only if there is a unique smallest population variance which is removed from the other variances. Such situations certainly exist, but the application of a procedure based on such a property alone, would be restricted. On the other hand, (2.4) implies no restriction on the configuration of population variances. Furthermore it guarantees that the selected population has variance close to the minimum of the population variances.

The sequential procedure for the construction of a confidence interval for μ with length at most $2d$, is then defined as follows:

1. Take r observations from each population and choose population π_ℓ where ℓ is defined by $S_\ell(r) = \min(S_1(r), \dots, S_k(r))$. Here r is determined for given P^* and Δ^* so that (2.3) (and 2.4)) is satisfied for the Bechhofer-Sobel procedure.

2. Let $N = \inf\{n \geq r : n \geq z^2 \hat{\theta}_\ell(n)/d^2\}$ where $\hat{\theta}_\ell(n)$ is defined as in (2.1) and (2.2). Let the resulting interval be $\hat{\mu}(N) \pm d$. Here $\hat{\mu}(N)$ is defined as in (2.1) where n_i for $i \neq \ell$ is replaced by r and n_i for $i = \ell$ is replaced in N . Also z is defined by $\Phi(z) = 1 - \beta/2$ and Φ is the $N(0, 1)$ distribution function.

3. ASYMPTOTIC PROPERTIES OF THE SEQUENTIAL PROCEDURE

Theorem 1

As $d \rightarrow 0$

$$(3.1) \quad N \rightarrow \text{a.s. and } EN \rightarrow \infty$$

$$(3.2) \quad \lim N d^2 / z^2 = \theta_\ell \text{ a.s.}$$

$$(3.3) \quad \lim EN d^2 / z^2 = E\theta_\ell = \sum_{i=1}^k \theta_i P(\ell = i)$$

$$(3.4) \quad \liminf P(|\hat{\mu}(N) - \mu| \leq d) \geq P^*(1 - \beta)$$

Proof

(3.1) and (3.2) follow using an argument similar to that in Lemma 1 of

Chow and Robbins (1965). Note that since $\hat{\theta}_i(n) \rightarrow \theta_i$ a.s. as $n \rightarrow \infty$ for each $i = 1, \dots, k$, we also have $\hat{\theta}_\ell(N) \rightarrow \theta_\ell$ a.s. as $d \rightarrow 0$. For (3.3) we first consider the uniform integrability of $\{Nd^2\}$.

$$\begin{aligned}
 P(Nd^2 > m) &\leq P(\hat{\theta}_\ell(t) > td^2/z^2) \text{ where } t = [m/d^2] \\
 &\leq P(\hat{\theta}_\ell(t) - \theta_\ell > h(m)) \text{ where } h(m) = m/z^2 - d^2/z^2 - \theta_p \\
 &\quad \text{and } \theta_p = \max(\theta_1, \dots, \theta_k) \\
 &\leq E(\hat{\theta}_\ell(t) - \theta_\ell)^2/h(m)^2 \\
 &\leq \sum_{i=1}^k E(\hat{\theta}_i(t) - \theta_i)^2/h(m)^2 \\
 &\leq C/m^2 \text{ for } C \text{ a constant and } m \text{ large.}
 \end{aligned}$$

Then Lemma 3.2 of Bickel and Yahav (1968) implies uniform integrability. Using the uniform integrability and (3.2) we then obtain $ENd^2 \rightarrow z^2 E\theta_\ell$ as $d \rightarrow 0$.

To prove (3.4) we make use of results on mixing limit theory. Theorem 8 of Csörgö and Fischler (1973) implies that $N^{\frac{1}{2}}(\bar{X}_i(N) - \mu)/\theta_i^{\frac{1}{2}} (i = 1, \dots, k)$ is mixing with limiting distribution Φ . We also use an analogue of Slutsky's theorem for mixing sequences (cf. Aldous and Eagleson (1978), theorem 1). Let $\Lambda = \{i_1, \dots, i_s\} = \{i : \theta_i/\theta_\lambda < \Delta^*, i = 1, \dots, k\}$. Then

$$\begin{aligned}
 &\liminf P(|\hat{\mu}(N) - \mu| \leq d) \\
 &\geq \liminf \sum_{j=1}^s P(|\hat{\mu}(N) - \mu| \leq d, \ell = i_j) \\
 &\geq \sum_{j=1}^s \liminf P(N^{\frac{1}{2}}|\bar{X}_\ell(N) - \hat{\mu}(N)|/\theta_{i_j}^{\frac{1}{2}} + N^{\frac{1}{2}}|\bar{X}_\ell(N) \\
 &\quad - \mu|/\theta_{i_j}^{\frac{1}{2}} \leq N^{\frac{1}{2}}d/\theta_\ell, \ell = i_j) \\
 &= \sum_{j=1}^s \Phi(z)P(\ell = i_j) \\
 &= (1 - \beta)P(\Lambda) \\
 &\geq (1 - \beta)P^*.
 \end{aligned}$$

The asymptotic results in theorem 1 are valuable, because the asymptotic coverage probability is established in (3.4). The asymptotic expression for EN is also useful, but does not help to give a picture of the total number of observations needed. (Denote the total number of observations, including those from the populations not selected after the first stage, by T . Then

$$T = (k-1)r + N, \text{ so that } Td^2 = (k-1)rd^2 + Nd^2$$

and $\lim ETd^2 = \lim ENd^2$). For this reason, we next consider an alternative asymptotic approach. Suppose that both β and P^* are functions of a quantity α such that both $\beta \rightarrow 0$ and $P^* \rightarrow 1$ as $\alpha \rightarrow 0$. Since r depends on P^* and N depends on β , this will lead to an asymptotic expression for ET which incorporates r . For mathematical convenience, we consider the case where $\hat{\theta}_t(n)$ is replaced in the stopping rule by $S_t(n)$.

Theorem 2

Let $N = \inf\{n \geq r : n \geq z^2 S_t(n)/d^2\}$. Then $z \rightarrow \infty$ and $P^* \rightarrow 1$ as $\alpha \rightarrow 0$ imply

$$(3.5) \quad \limsup EN/z^2 \leq \Delta^* \theta_\lambda/d^2.$$

In the case when $\theta_i/\theta_\lambda \geq \Delta^*$ for $i \neq \lambda$,

$$(3.6) \quad \lim EN/z^2 = \theta_\lambda/d^2.$$

Proof

The proof of (3.5) uses ideas similar to those in Perng (1969).

Let $\Omega_0 = \{\omega : \ell \in \Lambda\}$, where $\Lambda = \{i_1, \dots, i_s\}$ is as in the proof of Theorem

1. We first show that, as $\alpha \rightarrow 0$,

$$(3.7) \quad \int_{\Omega_0^c} N/z^2 \rightarrow 0$$

Note that

$$(3.8) \quad N/z^2 < 1/z^2 + S_\ell(N-1)/d^2, \text{ and}$$

$$\begin{aligned} S_\ell(N-1) &\leq \sum_{i=1}^k S_i(N-1) \\ &\leq \sum_{i=1}^k (N-1)^{-1} \sum_{j=1}^{N-1} (X_{ij} - \mu)^2 \\ &\leq \sum_{i=1}^k \sup_n n^{-1} \sum_{j=1}^n Y_{ij}, \text{ where } Y_{ij} = (X_{ij} - \mu)^2 \\ &= R \text{ (say), where } ER < \infty \end{aligned}$$

by the Marcinkiewicz-Zygmund inequality (cf. Chow and Teicher (1978) p.357).

This implies (3.7) by the Dominated Convergence Theorem and since $P(\Omega_0^c) \leq 1 - P^* \rightarrow 0$. Since $S_{i_j}(n) \rightarrow \theta_{i_j}$ as $n \rightarrow \infty$ for each $i_j \in \Lambda$, we get by Egoroff's theorem that, for given $\delta > 0$, there is a set A_{i_j} such that the convergence is uniform on A_{i_j} , and $P(A_{i_j}) > 1 - \delta$. This implies that, for given $\epsilon > 0$, there is a $n_0(\delta, \epsilon)$ such that

$$S_{i_j}(n) \leq \theta_{i_j} + \epsilon$$

for $n \geq n_0(\delta, \epsilon)$. Let $\Omega_1 = \{\omega : N \geq n_0(\delta, \epsilon) + 1\}$.

Using (3.8), we obtain

$$\begin{aligned} \int_{\Omega_0} N/z^2 &= \int_{\Omega_0 \cap \Omega_1} N/z^2 + \int_{\Omega_0 \cap \Omega_1^c} N/z^2 \\ &\leq P(\Omega_0 \cap \Omega_1)/z^2 + \sum_{j=1}^S \int_{\{\ell=i_j\} \cap \Omega_1} S_{i_j}(N-1)/d^2 + n_0(\delta, \epsilon)/z^2 \end{aligned}$$

But

$$\begin{aligned} \int_{\{\ell=i_j\} \cap \Omega_1 \cap A_{i_j}} S_{i_j}(N-1) &\leq (\theta_{i_j} + \epsilon) P(\{\ell = i_j\} \cap \Omega_1 \cap A_{i_j}) \\ &\leq \Delta^* \theta_\lambda P(\ell = i_j) + \epsilon \end{aligned}$$

and

$$\int_{\{i=i_j\} \cap \Omega_1 \cap A_{i_j}^c} S_{i_j}(N-1) \leq \int_{A_{i_j}^c} R.$$

Hence, as $\alpha \rightarrow 0$,

$$\limsup \int_{\Omega_0} N/z^2 \leq \limsup \Delta^* \theta_\lambda P(\Omega_0)/d^2 + \epsilon/d^2 + \left(\sum_{j=1}^S \int_{A_{i_j}^c} R \right) / d^2$$

so that (letting $\epsilon \rightarrow 0, \delta \rightarrow 0$)

$$\limsup \int_{\Omega_0} N/z^2 \leq \Delta^* \theta_\lambda / d^2.$$

The proof of (3.6) follows in a similar way.

Suppose now for example, that $P^* = (1 - \alpha)^{\frac{1}{2}}$ and $1 - \beta = (1 - \alpha)^{\frac{1}{2}}$, then it follows immediately from (3.4) that

$$(3.9) \quad \liminf P(|\hat{\mu}(N) - \mu| \leq d) \geq 1 - \alpha \text{ as } d \rightarrow 0.$$

In order to get an approximate expression for r as $\alpha \rightarrow 0$, we argue as follows: the requirement (2.3) determines r . But

$$\begin{aligned} P(\ell = \lambda) &= P(Y_i - Y_\lambda > -\log(\theta_i/\theta_\lambda), i = 1, \dots, k, i \neq \lambda) \\ &\quad \text{where } Y_i = \log(S_i(r)/\theta_i), i = 1, \dots, k \\ &= P(Z_i - Z_\lambda > -\log(\theta_i/\theta_\lambda)/(2/r)^{\frac{1}{2}}, i = 1, \dots, k, i \neq \lambda) \\ &\quad \text{where } Z_i = Y_i/(2/r)^{\frac{1}{2}} \text{ and is approximately} \\ &\quad N(0, 1) \text{ for } r \text{ large (cf. Bechhofer and Sobel (1954)).} \\ &\geq P(Z_i - Z_\lambda > h, i = 1, \dots, k, i \neq \lambda) \text{ provided} \\ &\quad \theta_i/\theta_\lambda \geq \Delta^* \text{ and where } h = (-\log \Delta^*)/(2/r)^{\frac{1}{2}} \\ &\approx \int_{-\infty}^{\infty} [1 - \Phi(z + h)]^{k-1} d\Phi(z). \end{aligned}$$

Since $h^2 \sim -4\log(1 - P^*)$ (cf. Bechhofer, Kiefer and Sobel (1968, p.185)), we find the following asymptotic expression:

$$(3.10) \quad r \sim -8\log\alpha/(\log\Delta^*)^2.$$

If (3.10) is accepted as asymptotic expression for τ , combination with (3.5) and (3.6) gives (note that $z^2 \sim -2\log\alpha$)

$$(3.11) \quad \limsup ET/(-\log\alpha) \leq 8(k-1)/(\log\Delta^*)^2 + 2\theta_\lambda\Delta^*/d^2 \text{ as } \alpha \rightarrow 0$$

while, if $\theta_i/\theta_\lambda \leq \Delta^*$.

$$(3.12) \quad \lim ET/(-\log\alpha) = 8(k-1)/(\log\Delta^*)^2 + 2\theta_\lambda/d^2 \text{ as } \alpha \rightarrow 0.$$

4. ASYMPTOTIC EFFICIENCY OF THE SEQUENTIAL PROCEDURE AS COMPARED WITH TWO-STAGE PROCEDURES

Alam and Saxena (1984) propose two-stages procedures of the following form. In the first stage a random sample of size $m(\geq 2)$ is taken from each population. In the second stage, observations are chosen from π_ℓ only, where ℓ is defined by $S_\ell(m) = \min(S_1(m), \dots, S_k(m))$. The size of the second sample is $M - m$, where

$$(4.1) \quad M = \max\{m, [\eta^2 S_\ell(m)/d^2] + 1\}$$

where $[x]$ denotes the largest integer less than or equal to x and η is such that the resulting confidence interval I satisfies $P(\mu \in I) \geq 1 - \alpha$. The total sample size is $T' = (k-1)m + M$. Two procedures are given, with associated constants η_t and η_b , sample sizes M_t and M_b and intervals I_t and I_b . The total sample sizes are $T'_t = (k-1)m + M_t$ and $T'_b = (k-1)m + M_b$. Then, if $E(S_\ell(m))$ is denoted by τ , it follows from (4.1) that

$$(4.2) \quad \lim ET'd^2 = \eta^2\tau \text{ as } d \rightarrow 0 \text{ and}$$

$$(4.3) \quad \lim ET'/\eta^2 = \tau/d^2 \text{ as } \alpha \rightarrow 0.$$

The constant η_t is defined by

$$(4.4) \quad (2F_{m-1}(\eta_t) - 1)^k = 1 - \alpha$$

where F_m is the distribution function of a Student t distribution with m degrees of freedom. We now derive an asymptotic expression for η_t as $\alpha \rightarrow 0$.

Lemma 1

Let η_t be defined by (4.4) then $\eta_t \sim C\alpha^{-1/m}$ as $\alpha \rightarrow 0$, where C depends on k and m only.

Proof

Note that (4.4) implies $\eta_t \rightarrow \infty$ as $\alpha \rightarrow 0$. Also $1 - (2F_{m-1}(\eta) - 1)^k \sim 2k(1 - F_{m-1}(\eta))$ as $\eta \rightarrow \infty$.

$$\begin{aligned} \text{But } 1 - F_{m-1}(\eta) &= C \int_{\eta}^{\infty} (1 + y^2/m)^{(m+1)/2} dy \\ &= C \int_0^w x^{m-1} (1 - x^2)^{-\frac{1}{2}} dx \text{ after substituting} \\ &\quad x = (1 + y^2/m)^{-\frac{1}{2}} \text{ and where } w = (1 + \eta^2/m)^{-\frac{1}{2}} \\ &\quad \text{cf. Johnson and Kotz (1970, p.112).} \\ &= C[w^m(1 - w^2)^{-\frac{1}{2}} - \int_0^w x^{m+1}(1 - x^2)^{-\frac{3}{2}} dx]. \end{aligned}$$

Since $w \sim m^{\frac{1}{2}}\eta^{-1}$, we see that

$$\left| \int_0^w x^{m+1}(1 - x^2)^{-\frac{3}{2}} dx \right| \leq w^{m+2}(1 - w^2)^{-\frac{3}{2}} = O(\eta^{-(m+2)})$$

and hence $1 - F_{m-1}(\eta) \sim C\eta^{-m}$. Then the result follows from (4.4).

If (3.9) holds the sequential procedure may be compared with the two-stage procedures of Alam and Saxena (1984). The two-stage procedures have coverage probability at least $1 - \alpha$ nonasymptotically, but may be conservative, as is indicated by some of the Monte Carlo results of Alam and Saxena (1984). On the other hand, preliminary Monte Carlo studies indicate that the coverage probability of the sequential procedure achieves the nominal value even nonasymptotically.

We first consider the case when $d \rightarrow 0$.

From (3.3) we obtain, as $d \rightarrow 0$,

$$\begin{aligned} \lim ENd^2/z^2 &= \sum_{i \in \Lambda} \theta_i P(\ell = i) + \sum_{i \notin \Lambda} \theta_i P(\ell = i) \text{ where, as before,} \\ \Lambda &= \{i : \theta_i/\theta_\lambda < \Delta^*, i = 1, \dots, k\} \\ &\leq \Delta^* \theta_\lambda P(\ell \in \Lambda) + (1 - P^*) \sum_{i \notin \Lambda} \theta_i. \end{aligned}$$

Under the assumptions immediately preceding (3.9), we obtain

$$(4.5) \quad \limsup_{\alpha \rightarrow 0} \lim_{d \rightarrow 0} ENd^2/z^2 \leq \Delta^* \theta_\lambda$$

which also gives an asymptotic upper bound for ETd^2/z^2 .

Using (4.2) and (4.5) we see that the asymptotic efficiency of the sequential procedure versus the two-stage procedures, as first $d \rightarrow 0$ and then $\alpha \rightarrow 0$, is asymptotically at least

$$(4.6) \quad \frac{\eta^2 \tau}{z^2 \Delta^* \theta_\lambda}.$$

Since $z^2 \sim -2 \log \alpha$ and $\eta_i^2 \sim C \alpha^{-1/m}$ (Lemma 1), it follows that (4.6) tends to infinity as $\alpha \rightarrow 0$ when η is η_i . When η is η_b a similar result should hold. (Alam and Saxena (1984) mention that from numerical work it appears that $\eta_i \leq \eta_b$.)

Although this does not change the result just obtained, we mention that, if it is assumed that $\theta_i/\theta_\lambda \geq \Delta^*$ for $i \neq \lambda, i = 1, \dots, k$, then a slight adjustment to the argument preceding (4.5) shows that (4.5) may be improved to

$$\lim_{\alpha \rightarrow 0} \lim_{d \rightarrow 0} ENd^2/z^2 = \theta_\lambda.$$

We also consider the asymptotic efficiency as $\alpha \rightarrow 0$ (without first letting $d \rightarrow 0$). This limiting situation has the advantage that, in the case of the sequential procedure, the number of observations needed for the selection procedure also appears in the limiting expression for the total sample size. Then (3.11) or (3.12) shows that the conclusion is similar to that obtained above.

5. DISCUSSION

The results obtained in this paper show that a sequential procedure for estimation of the common mean of several normal populations by means of a fixed-width confidence interval can be successfully constructed. A comparison of the proposed sequential procedure with the two-stage procedures of Alam and Saxena (1984) show that, in an asymptotic sense, the sequential procedure is infinitely more efficient.

One could also consider the possibility to construct a sequential procedure which makes use of a different selection procedure. For instance, Hoel (1971) and Swanepoel and Venter (1975) discuss truncated sequential procedures which have the desirable property of eliminating inferior populations as the sampling proceeds. Alternatively one could use a procedure which has as goal selection of t populations with the smallest variances ($1 \leq t \leq k$) where t is predetermined (cf. Bechhofer and Sobel (1954)) or a procedure which aims at selecting a subset (of random size) of the populations containing the population with the smallest variance (cf. Gupta and Sobel 1962)).

A final remark is that the normality assumption is needed in the asymptotic analysis only as far as the existence of certain moments is concerned. However, the selection procedure used is designed for the normal case.

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